Exact Periods and Labelled Orbit Counts for Closed Pointer Reversal

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Abstract

The familiar pointer-reversal assignment becomes a finite permutation when both registers and every stored pointer range over the same finite label set, with no sentinel or stopping rule. We classify its orbits on the full carrier. The active augmented-edge two-core is a figure-eight, barbell or theta, while all complementary pointers remain fixed. Explicit branch return states give seven exact-period cases, including loops and indistinguishable parallel edges. On $n \geq 2$ labels the attained periods are precisely $\{1, \ldots, 2n\} \cup \{2n+2, 2n+4, \ldots, 4n-4\}$. Observable orientation and order decorations then give an explicit rational bivariate core generating function and a labelled orbit census after arbitrary frozen complements are attached. We distinguish these closed-carrier results from the established reversal kernel, initialized-list analyses and fixed-digraph rotor theorems.

1 The closed carrier and its active core

For $n \geq 1$ put $V = [n]$ and $X_n = V^2 \times V^V$. On the full labelled carrier, with ordered registers, define

$$R(u, v, f) = (v, f(v), f[v := u]). \tag{1}$$

Every expression on the right uses the old state. We allow $u = v$, loops and arbitrary total maps f; there is no nil value, stopping rule or hidden edge identity. We determine two kinds of information: every exact period, and the orbit decorations needed for the labelled census. They share a return-map analysis, but a scalar period does not determine the decorations.

The kernel is established in-place list reversal, as in Manna and Waldinger [4]. Loginov, Reps and Sagiv [3] analyze a NULL-initialized, NULL-stopped version on possibly cyclic lists, including restored handles. Berdine et al. explicitly use the decreasing quantity $2i + j + k$ in their panhandle example [1, Example 8]. Thus neither reversal nor doubled-handle timing is claimed as new.

Rotor routing also has a complete established orbit theory. On a fixed finite strongly connected directed multigraph, with prescribed cyclic orders, let $T_G(v)$ count inward spanning arborescences and $M = \gcd_v T_G(v)$. Pham's theorem gives M recurrent rotor orbits, each of size $\sum_v d^+(v)T_G(v)/M$ [5, Theorem 1]; the size is independent of the chosen cyclic orders. The Eulerian tour case is also treated in Holroyd et al. [2, Lemma 4.9]. These are not our contribution. A barbell with cycle lengths $3, 3$ and bridge length 1 has pointer period 16 below, whereas its bidirected Eulerian rotor graph has 14 arcs and recurrent period 14. This excludes only a time-preserving identification on that same core, not factors, suspensions, enlarged states or arbitrary rotor encodings.

Some elementary facts fix the setting. The inverse is

$$R^{-1}(a, b, g) = (g(a), a, g[a := b]). \tag{2}$$

Substitution proves both identities, including coincident registers. Hence every state is periodic. A fixed state has $u = v = f(v)$; the other pointers are arbitrary, giving n^n fixed states. These facts receive no separate contribution claim.

Associate to (u, v, f) the undirected multigraph G with the stored edges $\{x, f(x)\}$ for all x, and one extra edge $\{u, v\}$. Edges have multiplicity but no identities; a loop contributes two to degree. Represent the extra edge by $u \to v$ and the other edges by their stored arrows. Equation (1) exchanges the stored and extra roles of two edges and reverses the traversed extra arrow, so G is invariant.

Lemma 1. *The component containing the registers has a connected two-core K with s vertices, $s+1$ edges and minimum degree two. Both registers stay in K. Every pointer outside K is frozen. The core is a figure-eight with cycle lengths a, b, a barbell with cycle lengths a, b and bridge length c, or a theta with path lengths a, b, c, all lengths positive. Their sizes are respectively $a+b-1, a+b+c-1, a+b+c-1$.*

Proof. Each functional component has equally many edges and vertices. Adding the register edge either augments one such component or joins two; in both cases the active component has one excess edge. The active vertex u has two outgoing arrows, and every other vertex has one. A degree-one vertex is therefore not u, and its single edge is its stored arrow directed toward the remaining graph. Removing it removes no remaining vertex's outgoing arrow. Induction through leaf pruning leaves u, the extra edge, and hence v in the two-core; all removed arrows point inward. Invariance of G makes those arrows, and all inactive components, frozen for all time.

The core retains excess one, so $\sum_{x \in K}(\deg x - 2) = 2$. There is one degree-four vertex or two degree-three vertices, all others having degree two. Following chains between branch incidences gives two closed chains in the first case; in the second it gives either three joining chains, or one joining chain and a closed chain at each end. These are exactly the asserted three shapes, with loops and length-two cycles included. □

2 Exact returns and observable decorations

Fix a labelled core K and all complementary pointers. A degree-two chain is forced: on arrival at an internal vertex, its old stored arrow becomes the next extra arrow and the incoming edge is stored in reverse. Induction traverses the chain and reverses it. At a branch the same rule chooses the old stored arrow and stores the reversed incoming edge. The outdegree conditions leave no additional internal choices. A rooted cycle has one observable orientation for length 1 or 2, and two for larger lengths: on a double edge, exchanging the two edges does not change any destination.

Theorem 2. *For every fixed labelled core and fixed complementary map, all periods and the number of distinct R-orbits are given in Table 1.*

Proof. Figure-eight. Let A be the common branch. Apart from the double loop, choose a prescribed cycle C_1 using its labels (the other is C_2). At an anchor with $u = A$ and the extra arrow starting C_1, the stored arrow at A starts C_2. The forced traversals are $C_1, C_2, C_1^{-1}, C_2^{-1}$. After $h = a + b$ steps, the anchor is restored with both cycle orientations reversed. If both reversals are invisible, the state returns at h. No earlier return is possible: branch departures

Core and degeneration	Exact period	Orbits
Figure-eight, $a = b = 1$	1	1
Figure-eight, $a, b \leq 2$, not both 1	$a + b$	1
Figure-eight, $\max(a, b) \geq 3$	$2(a + b)$	$1 + \mathbf{1}_{a,b\geq 3}$
Barbell, $a, b \leq 2$	$a + b + 2c$	1
Barbell, $\max(a, b) \geq 3$	$2(a + b + 2c)$	$1 + \mathbf{1}_{a,b\geq 3}$
Theta, $a = b = c = 1$	2	1
Any other theta	$2(a + b + c)$	1 if two paths direct; 2 otherwise

Table 1: All lengths are positive. Parallel edges have no state identities. In the last row, "two paths direct" means at least two.

alternate between the two cycles, whose first destinations differ unless both are loops. The double loop has a single, fixed state.

If a cycle has length at least three, choose an internal vertex z on it. In the full $2h$ block, $u = z$ occurs twice, with distinct next neighbours. One prescribed ordered pair ($u = z, v =$ chosen neighbour) occurs only once, so the minimal period is $2h$. This also rules out a shorter period at any other phase. At the prescribed-cycle anchor, states are exactly pairs of observable cycle orientations; the return acts by joint reversal. There are two classes exactly when both cycles have two orientations, and one otherwise. A state on an internal chain reaches a branch and then this anchor, proving exhaustion rather than inferring orbit counts from a state total.

Barbell. Choose the branch A by label and let B be the other. Anchor at the extra bridge departure from A toward B. The bridge is directed toward B, and the stored outgoing arrow at each branch lies on its own cycle; the two cycle orientations are the only choices. The forced block is bridge, B-cycle, reverse bridge, A-cycle. It has length $h = a + b + 2c$, restores the bridge anchor and reverses both cycles. That ordered departure from A occurs only at block boundaries: its bridge neighbour differs from any neighbour on the A-cycle, also for loops and double edges. Thus the exact period is h if both cycle reversals are invisible, and $2h$ otherwise. Joint reversal gives the same decoration count as above. From a cycle one reaches its branch and then the bridge; from the bridge one reaches a cycle and subsequently the chosen anchor. This covers every state.

Theta. Temporarily distinguish the three chains only for the proof. At an active branch their roles are outgoing extra, outgoing stored, and incoming, say (i, j, k). Traversing i gives roles (k, i, j) at the opposite branch: i reverses and the old outgoing chain there was k. Six traversals use i, k, j, i, k, j, once in each direction per chain, and restore the state after $2(a + b + c)$ steps. If a chain has an internal vertex z, one prescribed ordered departure from z occurs exactly once in that block, proving minimality even when the other two chains are direct. If all chains are direct, their temporary identities must be forgotten: $f(A) = B, f(B) = A$ throughout, and just two register states alternate.

At a fixed branch, every two traversals rotate the role triple. Three distinguishable chains give exactly two cyclic orders. Only direct chains can be identical, since every other chain has its own nonempty labelled internal set. Two or three direct chains merge the two orders into one multiset order. The oriented roles determine an anchor state, and forced chain traversal brings every state to a branch. These order classes are therefore precisely the orbits. $\square$

3 The complete period set

Corollary 3. *At $n = 1$ the sole state is fixed. For $n \geq 2$ the attained periods are*

$$\{1, \ldots, 2n\} \;\cup\; \{2n+2, 2n+4, \ldots, 4n-4\}, \tag{3}$$

where a reversed range is empty. The maximum is $4n - 4$.

Proof. Write $s \leq n$ for core size. Odd periods beyond 1 come from short-cycle rows: the short figure-eight gives 3, and a short barbell has $p = 2s + 2 - (a + b) \leq 2s$, with odd value $2s - 1$. A nonshort figure-eight or nontrivial theta has even period $2s + 2 \leq 4n - 4$ for $s \geq 3$. A nonshort barbell has $a + b \geq 4$ and $p = 4s + 4 - 2(a + b) \leq 4s - 4$. The $n = 2$ cases follow directly from the table.

Periods $1, 2, 3$ arise from a double loop, triple-direct theta, and figure-eight $(1, 2)$. For $2 \leq k \leq n$, a barbell $(a, b, c) = (1, 1, k - 1)$ has $s = k$ and period $2k$. For $2 \leq k \leq n - 1$, the barbell $(1, 2, k - 1)$ has $s = k + 1$ and period $2k + 1$. These give all values through $2n$. A theta $(1, 1, s - 1)$, $3 \leq s \leq n$, realizes half-periods $4, \ldots, n + 1$. For each $6 \leq k \leq 2n - 2$, put $c = \max(1, k - n - 1)$ and $r = k - 2c$. Then $r \geq 4$ and $s = r + c - 1 \leq n$: if $k \leq n + 2$, $s = k - 2$; otherwise $s = n$ and $r = 2n + 2 - k \geq 4$. The barbell $(1, r - 1, c)$ realizes period $2k$. Together these cover every remaining even value; the small values of n are included by the stated empty ranges. Pad any smaller core by inactive fixed loops. For sharpness use the two-loop barbell at $n = 2$, theta $(1, 1, 2)$ at $n = 3$, and barbell $(1, 3, n - 3)$ for $n \geq 4$. □

4 A rational labelled orbit census

Let $c_{s,p}$ be the number of orbits on all cores with a fixed s-element label set and exact period p, divided by $s!$, and let $C(t, q) = \sum_{s,p} c_{s,p} t^s q^p$. Thus t is an exponential-generating variable for labels and q marks the actual period, not an upper bound. Write $(n)_s = n!/(n - s)!$.

Theorem 4. *Set $x = tq^2$, $Q = x/(1 - x)$ and $D = (1 - x)^{-2} - (1 + x)^2$. Then*

$$C_8(t, q) = tq + t^2q^3 + \tfrac{1}{2}t^3q^4 + \tfrac{1}{4}tq^4D, \tag{4}$$

$$C_B(t, q) = \frac{t^2q^4(1 + tq)^2}{2(1 - tq^2)} + \frac{t^2q^8D}{4(1 - tq^4)}, \tag{5}$$

$$C_\Theta(t, q) = \tfrac{1}{2}t^2q^2 + \tfrac{1}{2}t^2q^6\left(Q + Q^2 + \tfrac{1}{3}Q^3\right), \tag{6}$$

$$C(t, q) = C_8(t, q) + C_B(t, q) + C_\Theta(t, q). \tag{7}$$

The number $o(n, p)$ of full-carrier orbits of exact period p is

$$o(n, p) = \sum_{s=1}^{n} (n)_s\, n^{n-s}[t^sq^p]C(t, q). \tag{8}$$

Moreover

$$C(t, 1) = \frac{t - t^2 + t^4/2 - t^5/12}{(1 - t)^3}, \qquad c_1 = 1,\; c_2 = 2,\; c_s = \frac{5s^2 + s + 24}{24} \quad (s \geq 3), \tag{9}$$

where $c_s = [t^s]C(t, 1)$. Consequently the total orbit count is the explicit sum $\sum_{s=1}^{n}(n)_s n^{n-s} c_s$.

Proof. An ordered list of ℓ internal labels contributes t^ℓ to a labelled EGF. A rooted undirected cycle of length a contributes $u_a t^{a-1}$, where $u_1 = u_2 = 1$ and $u_a = 1/2$ for $a \geq 3$: reversal acts freely on lists with at least two distinct labels, but not on empty or singleton lists. Multiplying by the orbit-decoration number in Theorem 2 gives

$$u_a u_b (1 + \mathbf{1}_{a,b\geq 3}) = \begin{cases} 1, & a, b \leq 2, \\ 1/2, & \max(a, b) \geq 3. \end{cases} \tag{10}$$

For a figure-eight, the shared root contributes t and the unordered cycle pair divides by 2, freely except for two loops. Add that exception as tq. The other short pairs give $t^2 q^3 + t^3 q^4/2$. All remaining ordered length pairs contribute $\frac{1}{4}\sum_{\max(a,b)\geq 3} t^{a+b-1} q^{2(a+b)} = tq^4 D/4$, proving (4). For a barbell, distinct branch labels make endpoint interchange free, again giving $1/2$; its bridge has $c - 1$ ordered internal labels. The short pairs contribute $\frac{1}{2}\sum_{a,b\leq 2, c\geq 1} t^{a+b+c-1} q^{a+b+2c}$, the first term of (5). The other pairs have coefficient $1/4$ by (10) and period $2a + 2b + 4c$; summing their geometric series gives the second term.

For a theta, choose an unordered pair of distinct branch labels, of weight $t^2/2$. Relative to either fixed ordering of those labels, a nondirect path is a nonempty ordered list of internal labels, with weight Q. Three direct paths contribute period weight q^2. If there are $k = 1, 2, 3$ nondirect paths, their unordered collection has weight $Q^k/k!$, and the decoration multipliers are $1, 2, 2$. The common remaining period weight is q^6. This yields $Q + Q^2 + Q^3/3$ in (6), including indistinguishable direct paths.

For (8), first choose the core label set S, then a core orbit decoration, and independently any map $g : V \setminus S \to V$. Such a map either follows trees into S or lies in inactive functional components. Adding it therefore creates no new active two-core and does not alter any core update. Conversely, leaf pruning a full state recovers S uniquely and recovers all values of g, which are frozen by Lemma 1. Thus this is a bijection on orbits, with $\binom{n}{s} s! n^{n-s}$ extensions per EGF coefficient, proving the formula.

For completeness, at $q = 1$ put $U = 1 + t + t^2/[2(1-t)]$, $W = t^2/[2(1-t)]$ and $Q = t/(1-t)$. The same cycle weights give

$$C_8 = \tfrac{t}{2}(U^2 + W^2 + 1), \qquad C_B = \frac{t^2(U^2 + W^2)}{2(1-t)}, \qquad C_\Theta = \tfrac{t^2}{2}(1 + Q + Q^2 + Q^3/3).$$

Their sum simplifies to (9). Using $[t^j](1-t)^{-3} = \binom{j+2}{2}$, with a negative index contributing zero, gives the displayed coefficients, including $s = 1, 2$ separately. □

The coefficients c_s need not be integers: they are normalized labelled counts. Period alone loses census information; on five fixed labels, figure-eights $(1, 5)$ and $(3, 3)$ both have period 12 but respectively one and two orbit decorations. As a consistency identity, not a third contribution, $[q\partial_q C(t, q)]_{q=1} = t(1 + 2t)/(1-t)^4$; after the same complement extension the period-weighted counts sum to $|X_n| = n^{n+2}$. The standard fixed-iterate count is $\sum_{p|k} p\, o(n, p)$.

5 Evidence scope and limitations

The arguments above apply to every n and require no finite experiment. An inherited candidate check covered all 4,356 states for $n = 1, 2, 3, 4$; it is separate from this manuscript's verification artifacts. The paper-local verifier has since been executed once initially and twice in a strict replay pair, each run making 72,476 named predicate checks in 46 classes over the same 4,356 states. All three 12,501,943-byte outputs agree byte for byte with the adopted canonical. A separate full-output semantic reconstruction made 12,375,789 primitive checks; for the identical pair its accepted evidence is reused under the unchanged dependency key, not rerun. These author-side checks do not constitute independent manuscript review or completion of build and visual gates. Two-long-cycle figure-eights first occur at core size 5, two-long-cycle barbells at 6, and theta cores with three nondirect paths at 5; these families are deductively covered, not experimentally exercised here. The conclusions concern exactly (1) on its full carrier. They neither establish global priority nor exclude alternative encodings into other finite dynamical systems.

References

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